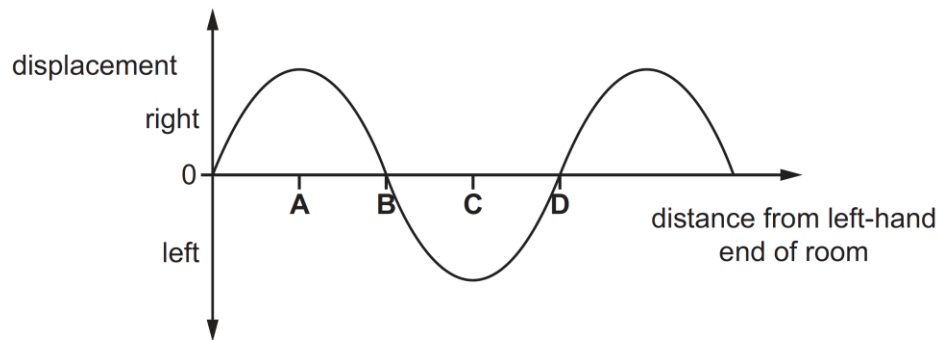


17

A sound wave travels from left to right across a room. The variation with distance across the room of the displacement of the air molecules at one instant is shown. At which distance will the air pressure be lowest?



Answer: D

The air molecule at A & the air molecules next to it are being displaced to the right at that instant in time.

Similarly, the air molecule at position C and the air molecules next to it are being displaced to the left at that instant in time.

The air molecule at B is not displaced but the air molecules next to it are being displaced towards it and the air pressure will be the highest there.

The air molecule at D is not displaced and the air molecules next to it are being displaced AWAY from it so the air pressure will be the lowest there.

[Turn over

18	Stationary sound waves are investigated using the apparatus shown
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